



1. What is the value of

$$4^1 - 3^2 + 2^3 - 1^4?$$

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

2. Let a , b , c , and d be real numbers that satisfy

$$a + 1 = b, \quad b + 2 = c, \quad c + 3 = d, \quad d + 4 = 21.$$

What is $a + b + c + d$?

(A) 51 (B) 52 (C) 53 (D) 54 (E) 55

3. Bill writes all odd perfect squares from 1 to 100, inclusive, and Jill writes all even perfect squares from 1 to 100, inclusive. Who writes more digits, and by how many?

(A) Bill,1 (B) Bill,2 (C) Jill,1 (D) Jill,2 (E) neither

4. Given a right triangle with legs of lengths 5 and 6, a square is drawn with one side as its hypotenuse such that the triangle is completely inside the square. What is the area of the region inside the square but outside the triangle?

(A) 46 (B) 47 (C) 48 (D) 49 (E) 50

5. A group of 200 people were invited to see a movie, where each viewer at the movie either has first row, second row, or third row seats. It is given that four-fifths of the people invited chose to watch the movie, one-ninth of the viewers were not invited, one-fifth of the viewers had first row seats, and 60 of the viewers had third row seats. What is the probability that a randomly selected viewer had second row seats?

(A) $\frac{3}{10}$ (B) $\frac{1}{3}$ (C) $\frac{7}{15}$ (D) $\frac{1}{2}$ (E) $\frac{4}{5}$

6. John is playing a game with 6 levels, each with 5 stages. After the third stage of each of the first five levels, John may choose whether or not to skip the remaining stages in the level and start at the first stage of the next level. If John finished the whole game, how many possible combinations of stages could John have played through?

(A) 5 (B) 10 (C) 16 (D) 30 (E) 32

7. What is the sum of all positive real numbers a such that the equation $x^2 + ax - 12 = 0$ has two distinct integer solutions x ?

(A) 6 (B) 12 (C) 14 (D) 16 (E) 22

8. Daniel has to walk one mile to complete his gym homework. He decides to split his path into quarters, where after each quarter, he randomly chooses to turn 90° clockwise or counterclockwise with equal probability. If Daniel walks in a straight line each quarter, what is the probability that he will end up where he started after walking the mile?

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{1}{2}$ (D) $\frac{5}{8}$ (E) $\frac{3}{4}$

9. Let a and b be positive integers. If a is divisible by 2 but not 3, and b is divisible by 3 but not 2, what is the greatest possible three-digit value of $a + b$?

(A) 995 (B) 996 (C) 997 (D) 998 (E) 999

10. How many orderings of the six numbers 1, 1, 2, 2, 3, and 6 are there such that the sum of the first three numbers is twice the sum of the last three numbers?

(A) 9 (B) 18 (C) 27 (D) 36 (E) 72

11. Given a triangle, a line is drawn such that it intersects the triangle in exactly two points. Which of the following statements must always be true?

- (A) The triangle is split into a smaller triangle and quadrilateral (B) The segment of the line in the triangle is shorter than every side of the triangle (C) At least two of the triangle's angle bisectors meet the line inside the triangle (D) The perimeters of each of the regions formed are less than that of the triangle (E) None of the above

12. In a plane, eight rays emanate from a point P such that every two adjacent rays form an acute angle with measure 45° . Next, a line segment with a finite length is drawn in the plane. If the line segment intersects exactly n of the rays, what is the sum of all possible values of n ? (If the line segment passes through P , then $n = 8$.)

- (A) 13 (B) 14 (C) 17 (D) 18 (E) 23

13. A positive integer is called *fresh* if the sum of the squares of its digits in base-4 is greater than the integer itself. How many fresh positive integers are there?

- (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

14. Draw two identical non-intersecting circles, a line tangent to both circles at distinct points A and B , where the circles are on the same side of the line, and a line tangent to both circles at distinct points C and D , where the circles are on opposite sides of the line. The lines intersect at point P . If $AB = 11$ and $CD = 5$, what is $AP \cdot BP$?

- (A) 20 (B) 22 (C) 24 (D) 26 (E) 28

15. For certain real numbers x and y , the first 3 terms of a geometric progression are $x - 2$, $2y$, and $x + 2$ in that order, and the sum of these terms is 4. What is the fifth term?

- (A) $\frac{64}{3}$ (B) $\frac{196}{9}$ (C) $\frac{512}{23}$ (D) $\frac{256}{11}$ (E) $\frac{128}{5}$

16. In the coordinate plane, let $P = (10, 0)$ and $Q = (a, a)$, for some real number a . It is given that when point P is reflected over point Q , the resulting point P' is 2 times as far from the origin as point P is. What is the sum of all possible values of a ?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

17. For positive integers n , let the n th triangular number be the sum of the first n positive integers. For how many integers n between 1 and 100, inclusive, does the n th triangular number have the same last digit as the product of the first n triangular numbers?

- (A) 11 (B) 12 (C) 20 (D) 21 (E) 22

18. Let x and y be real numbers such that

$$\begin{aligned} |x - |y - x|| &= 1 \\ |y - |x - y|| &= 2 \end{aligned}$$

What is the largest possible value of $x + y$?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

19. A car moves such that if there are n people in it, it moves at a constant rate of 4^n miles per hour. At noon, the car has 1 person in it and starts moving. After every mile, another person instantaneously gets in the car. How many people are in the car when the average speed the car has moved since noon reaches 17 miles per hour?

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

20. In rectangle $ABCD$, point A is reflected over diagonal $\overline{BD}$ to a point A' . If $A'B = A'C$ and $AA' = 6$, what is the area of rectangle $ABCD$?

- (A) 18 (B) $8\sqrt{6}$ (C) $12\sqrt{3}$ (D) 21 (E) $9\sqrt{6}$

21. Richard thinks of a positive integer n and writes the base ten representations of $n!$ and $(n+1)!$ on a board. He then erases the zeroes to the right of the last nonzero digit of each number (if any exist), resulting in two numbers a and b . If one of a and b is 4 times the other, what is the sum of all possible values of n less than 1000?

- (A) 315 (B) 441 (C) 656 (D) 714 (E) 819

22. In the xy -plane are perpendicular lines $y = ax + d$ and $y = bx + c$, where a , b , c , and d are real numbers in a geometric progression in that order. If the two lines and the line $y = \frac{3}{2}x$ pass through a common point, what is the least possible value of $a + b + c + d$?

$$(A) \frac{3}{2} \quad (B) \frac{51}{32} \quad (C) \frac{13}{8} \quad (D) \frac{111}{64} \quad (E) \frac{7}{4}$$

23. There are 15 people in a room, where everyone shakes hands with a positive number of other people in the room exactly once. If exactly 6 people shook 1 hand, exactly 5 people shook between 2 and 4 hands, inclusive, exactly 1 person shook 8 hands, and exactly 1 person shook 14 hands, what is the least possible total number of handshakes?

$$(A) 24 \quad (B) 25 \quad (C) 26 \quad (D) 27 \quad (E) 28$$

24. In right $\triangle ABC$ with $BC = 7$ and a right angle at A , let the midpoint of side $\overline{AB}$ be D . Suppose that there exists a point E on the circumference of the circumcircle of $\triangle ABC$ such that $\triangle CDE$ is equilateral. What is the side length of $\triangle CDE$?

$$(A) \sqrt{21} \quad (B) 2\sqrt{7} \quad (C) \frac{7\sqrt{21}}{6} \quad (D) \frac{7\sqrt{10}}{4} \quad (E) 3\sqrt{6}$$

25. Each of the six vertices of the regular hexagon shown below is labeled with either a 1 or a 2. Some diagonals of the hexagon are drawn, and each of the six points of intersection is labeled with either a 2, a 3, or a 4. In how many ways can the 12 points be labeled such that for every drawn diagonal of the hexagon, the sum of the numbers on its two endpoints is not equal to either of the numbers on the two points of intersection of the diagonal? Rotations and reflections are considered distinct.

$$(A) 502 \quad (B) 514 \quad (C) 526 \quad (D) 538 \quad (E) 550$$

Time limit: 75 minutes